

1. TONG QUAN HE THONG ROBOT & AN TOAN NHA MAY

Tai lieu nay la Huong dan Ky thuat Tong hop (Comprehensive Technical Manual) cho Robot DENSO RC8. He thong duoc trang bi bo dieu khien trung tam ECU 24V DC, cam bien nhiet do truc Axis 3 va cong truyen thong RS485 toc do cao.

CAC QUY TAC AN TOAN BAT BUOC:

- Khi phat hien ma loi E-404 (Motor Overheat) hoac E-500 (RS485 Timeout), lap tuc nhan nut Emergency Stop.
- Kiem tra dien ap cap truoc khi khoi dong lai: Dien ap dinh muc Supply Voltage = 24.0V DC.
- Sai so dien ap cho phat Tolerance trong khoang +/- 5% (22.8V - 25.2V DC).

CANH BAO AN TOAN (SAFETY ALERT):

Khi Nut Ngat Khan Cap Emergency Stop duoc kich hoac, toan bo r-le K1 se ngat nguon Servo.

2. SO DO MACH DIEN & CAP NOI WIRING DIAGRAM

SO DO MACH NGAT KHAN CAP EMERGENCY STOP & CHAN WIRING ECU:

[SCHEMATIC DIAGRAM 1]

Emergency Stop Switch (S1)

Relay K1 -> 24V DC Line

[ECU CONNECTOR J3 PINOUT]

Pin 1-2: 24V DC Power Input

Pin 3-4: RS485 TX/RX Signal

[SENSOR SCHEMATIC DIAGRAM - AXIS 3]

Vi tri Cam bien Nhiet do TH1 duoc dat tai dau noi Servo Motor Axis 3.

Nhiet do toi da cho phep: 85 C. Khi vuot qua 85 C, he thong se phat Ma loi E-404.

3. BANG THONG SO KY THUAT CONTROLLER (SPEC TABLE)

BANG MATRIX GIATI THONG SO DINH MUC THIET BI DENSO:

STT	Ten Thong So (Parameter)	Gia Tri Dinh Muc	Gioi Han An Toan	Trang Thai
1	Supply Voltage	24.0V DC	22.8V - 25.2V	NORMAL
2	Rated Current	5.5 Amp	Max 7.0A	NORMAL
3	Operating Temp	-10 C to +60 C	-15 C to +75 C	NORMAL
4	Motor Speed	3000 RPM	Max 3600 RPM	NORMAL
5	Voltage Tolerance	+/- 5%	Max +/- 8%	NORMAL
6	RS485 Baud Rate	115200 bps	9600-115200 bps	ACTIVE
7	Thermal Limit E-404	85.0 C	Max 90.0 C	PROTECTED

4. NHAT KY MA LOI & QUY TRINH KHOP LOI (LOGS & SOP)

NHAT KY LOI HE THONG FACTORY ERROR LOGS 2026:

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[2026-09-01 08:30:11] WARN - Voltage drop detected on 24V Line Drop: Supply voltage fell to 21.8V DC.  
[2026-09-01 09:12:45] ERROR - Code E-404: Motor Overheat Warning on Axis 3 Servo Motor. Temp: 87.5C.  
[2026-09-01 09:13:00] ACTION - Emergency Stop triggered automatically by Safety Relay K1.  
[2026-09-01 09:25:30] ERROR - Code E-500: RS485 communication timeout error between Mainboard & Pendant.  
[2026-09-01 10:05:12] WARN - Payload Exceeded: Manipulator arm overload warning during pick-and-place.  
[2026-09-01 10:30:00] RESOLVED - Reset Emergency state recovery completed. System restored.
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QUY TRINH KHAC PHUC SU CO (TROUBLESHOOTING SOP):

- 1. Khi xuất hiện lỗi E-404 Motor Overheat:
 - Kiểm tra nhiệt độ, có quá nhiệt? Axis 3 nhiệt độ > 50 C.
 - Kiểm tra quạt tản nhiệt và cảm biến nhiệt độ TH1 trên Pinout J3.
- 2. Khi xuất hiện lỗi E-500 RS485 Timeout:
 - Kiểm tra dây truyền thông Pin 3-4 trên ECU Controller.
 - Kiểm tra điện trở tải kháng 120 Ohm.
- 3. Khi ngừng khẩn cấp Emergency Stop:
 - Xoay nút reset ngừng khẩn cấp theo chỉ dẫn kim đồng hồ để khôi phục trạng thái Reset Emergency.